GENERAL CHEMISTRY

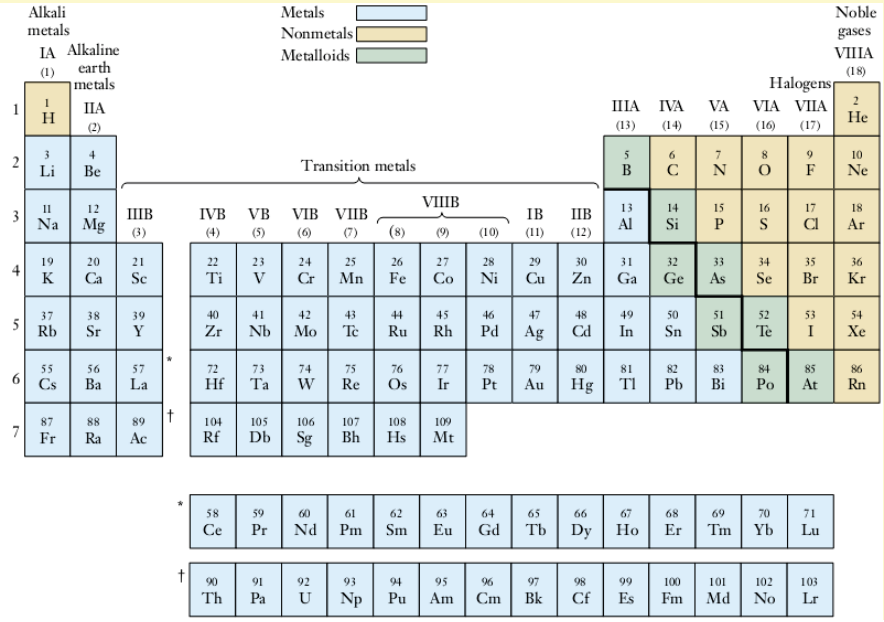
Chapter 2

Chemical periodicity

1

1

The Periodic Table



2

2

The Periodic Table

- **Columns** in the periodic table are called *groups* or families (IA, IIA, IIIB÷VIIIB, IB, IIB, IIIA ÷VIIIA).
- **Rows** in the periodic table are called *periods (n=1÷7, n is also the Principal quantum number)*.
- **Elements in a group** have similar chemical and physical properties and those within periods have properties that change progressively across the table.
- About **80%** of the elements are **metals**.

3

3

The Periodic Table

- Metals are located on the left hand side of the periodic table (most of the elements are metals).
- Non-metals are located in the top right hand side of the periodic table.
- Elements with properties similar to both metals and non-metals are called metalloids and are located at the interface between the metals and non-metals.

4

4

Properties of The Periodic Table

- Some of the groups in the periodic table are given special names.
- These names indicate the similarities between group members:
  - ✓ Group 1A: Alkali metals.
  - ✓ Group 2A: Alkaline earth metals.
  - ✓ Group 6A: Chalcogens.
  - ✓ Group 7A: Halogens.
  - ✓ Group 8A: Noble gases.

5

5

Properties of The Periodic Table

✓ **Period 1:** H ($1s^1$) & He ($1s^2$).

✓ **Periods 2 &3:** 8 elements in each period,

$2s^1 \rightarrow 2s^2 2p^6$, $3s^1 \rightarrow 3s^2 3p^6$

✓ **Periods 4 &5 :**18 elements in each period : 2 (s) + 10 (d) + 6 (p).

→ $2(ns^{1,2})$ + 10 $ns^2(n\text{-}1)d^{1\div10}$ +6 $(np^{1\div6})$

6

6

Properties of The Periodic Table

✓ **Period 6 :** 32 elements

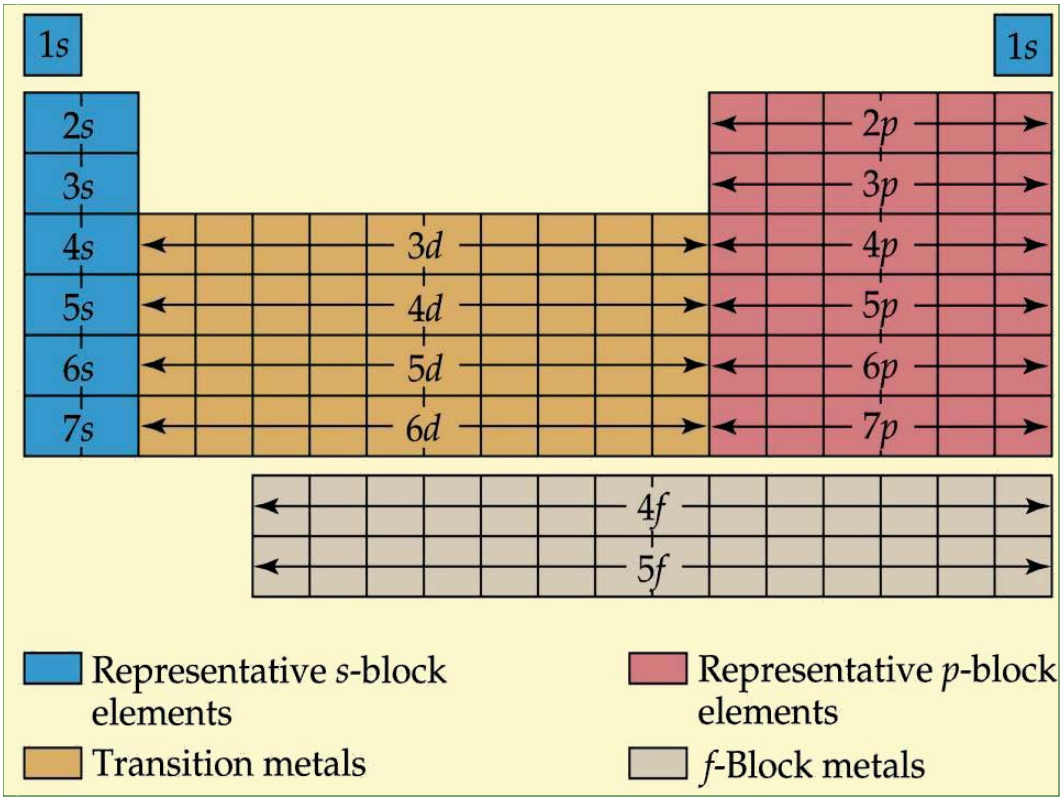
2(s)+14(f)+10(d)+6(p)

→ $2(6s^{1,2}) + 14\ (6s^2\ 4f^{1\div14})+10(6s^2\ 5d^{1\div10})+6\ (p^{1\div6})$

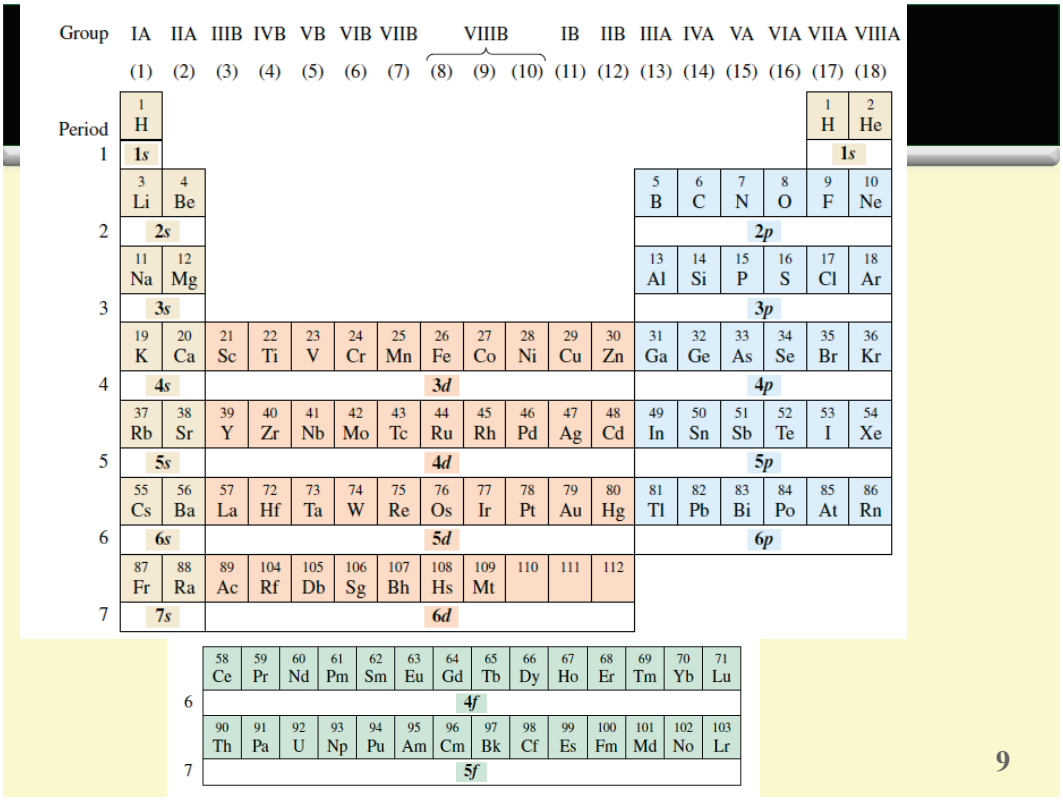
✓ **Period 7** : 31 elements (7 elements in A group, 10 d-elements and 14 Lathanit elements.

7

7



8



9

9

Properties of The Periodic Table

❖ THE A GROUPS ELEMENT (IA-VIIIA):

- ✓ The A groups contain elements in which s and p orbitals are being filled.
- ✓ Elements within any particular A group have similar electron configurations and chemical properties.
- ✓ *As the valence-shell configuration of ns^a or ns^2np^b that element is so called s-element or p-element, respectively.*

10

10

Properties of The Periodic Table

❖ **THE A GROUPS ELEMENT (IA-VIIIA):**

- ✓ Lithium, sodium, and potassium: have a single electron in their outermost sorbital (ns^1)- Group IA.
- ✓ Beryllium and magnesium have two electrons in their outermost shell, ns^2- Group IIA.
- ✓ Boron and aluminum have three electrons in their outermost shell, ns^2np^1- Group IIIA.

11

11

Properties of The Periodic Table

❖**THE A GROUPS ELEMENT (IA-VIIIA):**

- ✓ The valence-shell configuration of main group is ns^anp^b:
  - ➢ *n: the principle quantum number of the outer shell (the period number for the element).*
- ✓ *Group number = a+b.*

12

12

Properties of The Periodic Table

Group	s-, p-element
I A	ns^1
IIA	ns^2
IIIA	ns^2np^1
IVA	ns^2np^2
VA	ns^2np^3
VIA	ns^2np^4
VIIA	ns^2np^5
VIIIA	ns^2np^6

Example:

n=4, number of valance electron is 3.

Valance-shell configuration?

$4s^24p^1$

13

13

Properties of The Periodic Table

❖ **THE B GROUPS ELEMENT (IB-VIIIB):**

- ✓ The B groups are those in which there are one or two electrons in the s orbital of the outermost occupied shell, and the d orbitals, one shell smaller, are being filled.
- ✓ Example: $(n-1)d^1ns^2$, $(n-1)d^5ns^1$.
- ✓ Elements in the B groups are known as the d-transition elements or transition elements or transition metals.

14

14

Properties of The Periodic Table

Group	d-element
IB	$(n-1)d^{10}ns^1$
IIB	$(n-1)d^{10}ns^2$
IIIB	$(n-1)d^1ns^2$
IVB	$(n-1)d^2ns^2$
VB	$(n-1)d^3ns^2$
VIB	$(n-1)d^5ns^1$
VIIB	$(n-1)d^5ns^2$
VIIIB	$(n-1)d^{6,7,8}ns^2$

✓$ns^2(n-1)d^4 \rightarrow ns^1(n-1)d^5$

✓$ns^2(n-1)d^9 \rightarrow ns^1(n-1)d^{10}$

15

15

Properties of The Periodic Table

❖ **THE B GROUPS ELEMENT (IB-VIIIB):**

✓**f-Transition Elements**:

- *Electrons are being added to f orbitals.*
- *All are metals.*
- *The f-transition elements are located between Groups IIIB and IVB in the periodic table.*

16

16

Trends in The Periodic Table

1. Atomic radius
2. Ionic radius
3. Ionization energy
4. Electron affinity

17

17

Atomic Radius

- Two factors that determine the size of the outermost orbital: n and the effective nuclear charge.

$$Z_{eff} = Z - S$$

where Z_{eff} => effective nuclear charge,

Z => nuclear charge, atomic number

S => shielding constant.

Effective nuclear charge is the positive charge that an electron experiences from the nucleus, equal to nuclear charge but reduced by any shielding of screening from any intervening electron distribution.

18

18

Atomic Radius

- **Increase** top to bottom down a group

  The effective nuclear charge remains nearly constant, but n gets larger → the atomic radius increases.

- **Decrease** left to right across a period:

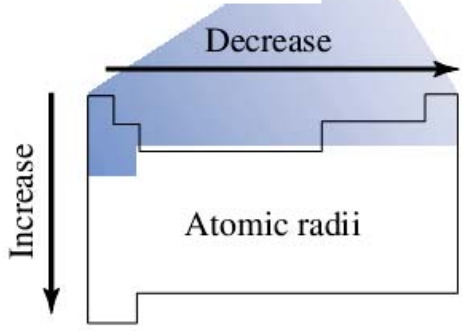
  Principal quantum number of outer orbitals remains constant. The effective nuclear charge increases, because the nuclear charge increases and the number of core electrons remain constant → pulling electrons towards the nucleus, decreasing size

- **Increases from upper right corner to the lower left corner**

19

19

Atomic Radius

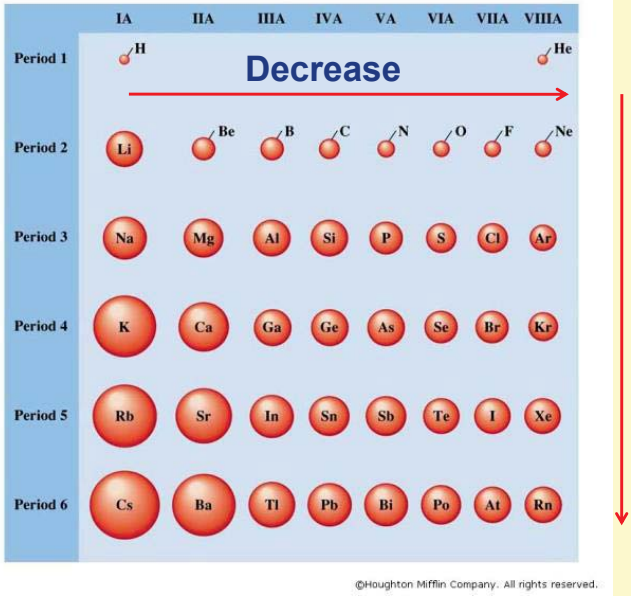


General trends in atomic radii of A group elements with position in the periodic table.

20

20

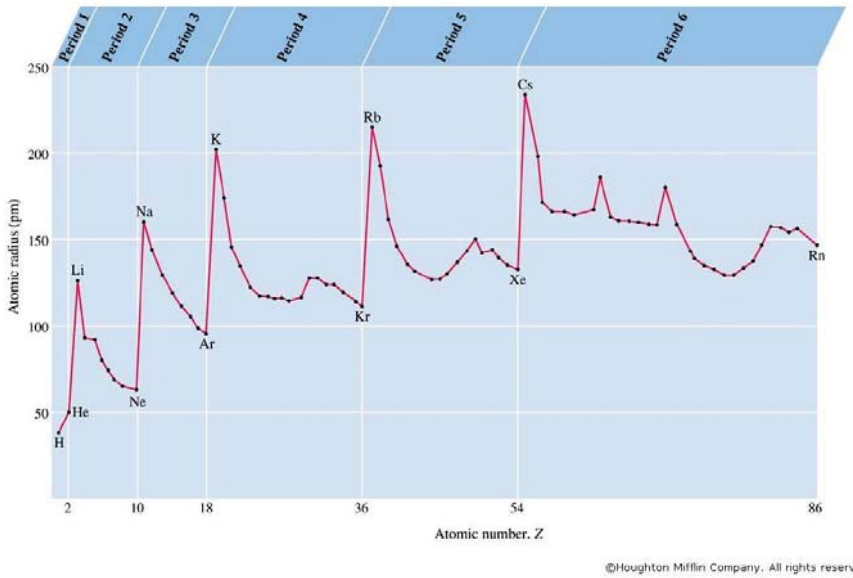
Atomic Radius



21

21

Atomic Radius vs. Atomic Number



22

22

Atomic Radius

Example:

Arrange the following elements in order of increasing atomic radii. Justify your order.

$_{55}Cs$, $_{9}F$, $_{19}K$, $_{17}Cl$

Sol.:

F and Cl are in group VIIA: F<Cl

Na and Cl are in period 3: Na>Cl.

Na, K and Cs are in group IA: Na<F<Cs

Thus:

F<Cl< K< Cs

23

23

Atomic Radius

Problem:

Arrange the following elements in order of increasing atomic radii. Justify your order.

a) Al (z=13), C (z=6), Si (z=14)

b) Na (z=11), Be (z=4), Mg (z=12)

Sol.:

C<Si<Al

Be<Mg<Na

24

24

Ionic Radius

- Same trends as for atomic radius
- Positive ions smaller than atom
- Negative ions larger than atom

$\Rightarrow r(A^+) < r(A) < r(A^-)$

25

25

Ionic Radius

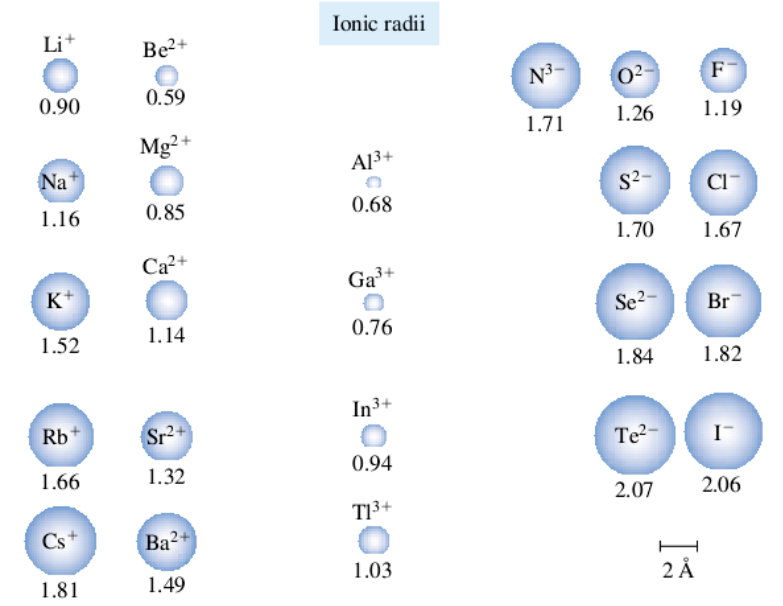
Isoelectronic Series

- Series of negative ions, noble gas atom, and positive ions with the same electronic configuration
- Size decreases as "positive charge" of the nucleus increases

26

26

Ionic Radius



27

27

Ionic Radius

EXAMPLE 6-4 ***Trends in Ionic Radii***

Arrange the following ions in order of increasing ionic radii: (a) Ca^{2+}, K^+, Al^{3+}; (b) Se^{2-}, Br^-, Te^{2-}.

$_{20}Ca$, $_{19}K$, $_{13}Al$, $_{34}Se$, $_{35}Br$, $_{52}Te$

Solution

(a) Ca^{2+} and K^+ are isoelectronic (18 electrons each) with an outer-shell electron configuration of $3s^23p^6$. Because Ca^{2+} has a higher nuclear charge (20+) than K^+ (19+), Ca^{2+} holds its 18 electrons more tightly, and Ca^{2+} is smaller than K^+. Al^{3+} has electrons only in the second main shell (outer-shell electron configuration of $2s^22p^6$), so it is smaller than either of the other two ions.

$$Al^{3+} < Ca^{2+} < K^+$$

$$Br^- < Se^{2-} < Te^{2-}$$

28

28

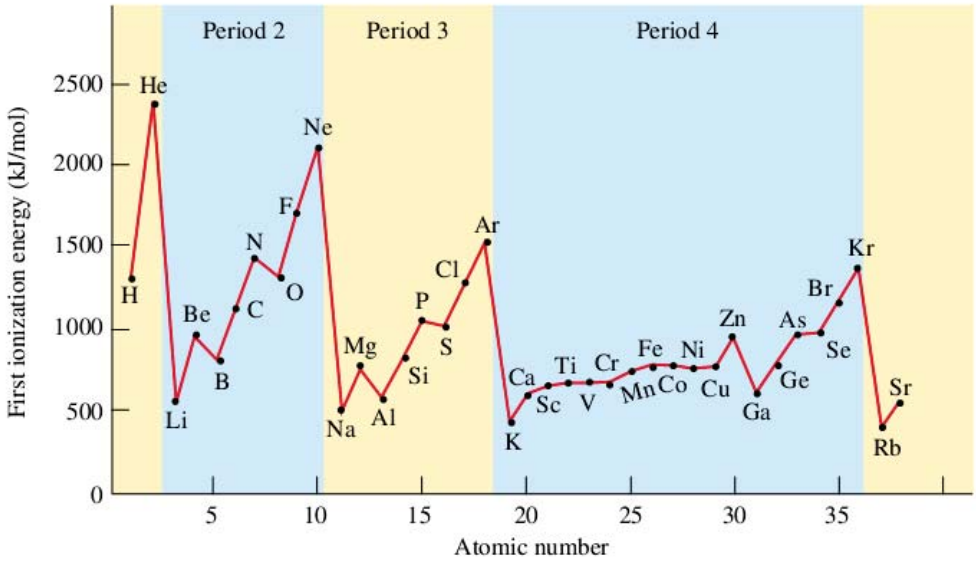
Ionization Energy

- The **first ionization energy (IE1)** of an atom is minimum energy needed to remove the highest-energy electron from the neutral atom in the gaseous state.
- In general, **within each period, IE1 value tend to increase with atomic number**.
- The **first ionization energy tend to decrease in going down any column of main- group elements.**

29

29

Ionization Energy vs. Atomic Number



30

Ionization Energy

- High value for non-metals, electrons difficult to remove.
- Increases from lower left corner of periodic table to the upper right corner.
- The second ionization energy (IE2) is the amount of energy required to remove the second electron. For given element, IE2 is always greater than IE1

31

31

Ionization Energy

Example:

$$Ca(g) + 590\ kJ \longrightarrow Ca^+(g) + e^-$$

$$Ca^+(g) + 1145\ kJ \longrightarrow Ca^{2+}(g) + e^-$$

$$Al(g) \longrightarrow Al^+(g) + e^- \qquad I_1 = 580\ kJ/mol$$
$$Al^+(g) \longrightarrow Al^{2+}(g) + e^- \qquad I_2 = 1815\ kJ/mol$$
$$Al^{2+}(g) \longrightarrow Al^{3+}(g) + e^- \qquad I_3 = 2740\ kJ/mol$$
$$Al^{3+}(g) \longrightarrow Al^{4+}(g) + e^- \qquad I_4 = 11{,}600\ kJ/mol$$

32

32

Ionization Energy

Deviations:

- A IIIA element (ns^2np^1) has smaller IE than the preceding IIA element (ns^2).

Apperently, the np electron of the IIIA element is more easily removed than one of the ns electron of the IIA element.

- A VIA element (ns^2np^4) has smaller IE than the preceding VA element (ns^2np^3).

As a result of electron repulsion, it is easier to remove an electron from the doubly occupied np orbital of the VIA element than from a singly occupied orbital of the VA element.

33

33

Ionization Energy

Example:

Arrange the following elements in order of increasing first ionization energy. Justify your order.

Na (z=11), Mg (z=12), Al (z=13), Si (z=14)

Sol.:

Si (Z=14), e^- configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

IE1 (Na)< IE1 (Mg), IE1 (Al) < IE1 (Si)

Al ($1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$) → IE1 (Al) < IE1 (Mg)

Thus:

IE1 (Na)< IE1 (Al)< IE1 (Mg)< IE1 (Si)

34

34

Ionization Energy

Exercise:

Arrange the members of each of the following elements in order of increasing first ionization energies:

$_{35}Br$, $_{9}F$, $_{31}Ga$, and $_{55}Cs$.

Cs<Ga<Br<F

35

35

Electron Affinity

- Energy released when an electron is added to an neutral atom.
- The **electron affinity (EA)** is the energy change for the process of adding an electron to the neutral atom in the gaseous state to form negative ion.
- If the negative ion is stable, the energy change for its formations is negative number.
- Large negative numbers indicate that a very stable negative ions formed.

36

36

Electron Affinity

- Same trends as ionization energy, increases from lower left corner to the upper right corner.
- Metals have low "EA".
- Nonmetals have high "EA".

37

37

Electron Affinity

EXAMPLE 6-3 ***Trends in EAs***

Arrange the following elements in order of increasing values of electron affinity, that is, from most negative to least negative.

K, Br, Cs, Cl

Solution

The order of increasing values of electron affinity is

(most negative EA) $Cl < Br < K < Cs$ (least negative EA)

38

38